

HS4002: Lecture 8C

Multivariate Analysis III

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Midterm (summary statistics)

- Mean = 65.
- SD = 20.

- Probability vs non-probability sampling.
- Total survey error framework
- Statistics, generally speaking.
 - Interpreting key statistics.
 - Reading regression tables.

What next?

- I'd encourage you to treat your test scores as a diagnostic.
- I'll be pulling up the exam + examples of responses I'm looking for on NTULearn for viewing.
- Come to office hours (or email me for appointment) if you want to review your own manuscript.
- I'll be releasing unstandardized 'RAW' test-scores so you get a clear sense of how you did.
 - As you know, scores may get normalized ('curved') at some point.

Count Model

Counts?

What should we do if we're interested in a dependent variable that takes the form of counts (e.g. the number of children a person has)?

We can continue to use the OLS:

$$E(y|x_1, x_2 \dots x_k) = \beta_0 + \beta_1 x_1 + \dots \beta_k x_k$$

where we can interpret the beta coefficients to reflect expected changes in counts of Y when there's a change in independent variable, *ceteris paribus*.

How to interpret results

If our dependent variable was no. of children,

- β_{income} would tell us the expected change in the number of children a person has when X_{income} increases by one unit, ceteris paribus.

Limitations with the OLS

- We can get predictions that are less than one.
- It doesn't really make sense to make a prediction of -1 children.

Poisson Regression

The Poisson Model is an example of a count model,

$$E(y|x_1...x_l) = \exp(\beta_o + ... \beta_k X_k)$$

where the exponential function, $\exp$, forces the model output to take on values that are non-negative. In a Poisson model, we assume that the nominal distribution underlying the data is a Poisson distribution.

We use maximum likelihood estimation to produce estimates of $\hat{\beta}$. The probability that a y equals the value of h can be expressed as,

$$P(y = h|\mathbf{x}) = \exp[-\exp(\mathbf{x}\beta)][\exp(\mathbf{x}\beta)]^h / h!$$

where $\mathbf{x}$ is a set of independent variables.

- You do not interpret it the same way as a linear OLS model.

- You can interpret β_k as an **increase in the log count**:
 - e.g. "A one unit change in x_k is expected to cause a β_k increase in the log count ($p < 0.01$), ceteris paribus."
- The beta coefficients indicate **factors of change**:
 - e.g. "A one unit change in x_k is expected to cause a change in y by a factor of $\exp(\beta_k)$ ($p < 0.01$), ceteris paribus."
- You can alternatively interpret this in terms of **percentage change**.
 - e.g. "A one unit change in x_k is expected to cause a $\{100 \times [\exp(\beta_k) - 1]\}\%$ increase in y ($p < 0.01$), ceteris paribus."

Negative Binomial Regression

Negative Binomial Regression

The Negative Binomial Model is an extension of the Poisson model that adds a parameter to account for overdispersion:

$$E(y|x_1...x_l) = \exp(\beta_o + ... \beta_k X_k + \epsilon).$$

The Poisson model makes the Poisson variance assumption that $Var(y|x) = E(y|x)$, which may not always be justified. When we think there is significant overdispersion (i.e. variance is significantly higher than what the Poisson variance assumption posits), we may choose to use the neg-bin model instead.

- The ϵ term above allows us to model for unobserved heterogeneity for any given combination of independent variables (under a Poisson model, they'd all have the same variance).
- Results from a neg-bin model are interpreted in the same way as those from a Poisson model.

Test for Overdispersion

- We can run likelihood ratio tests for overdispersion.
 - Recall that overdispersion in this case refers to the mean and variance of the dependent variable being equal.
- The restricted model in this case posits that the alpha value is zero (i.e. it is a Poisson model).
- If we reject the null, then there is overdispersion and we may want to use neg-bin models over Poisson models.
- If we accept the null, we think alpha may well be zero, and we may want to use Poisson models over neg-bin models.

Interpreting $\hat{\beta}$ from negbin models

- Exactly the same as Poisson models.

- You can interpret β_k as an **increase in the log count**:
 - e.g. "A one unit change in x_k is expected to cause a β_k increase in the log count ($p < 0.01$), ceteris paribus."
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Comparing Models

- We can use the same AIC statistic as with OLS regression.
 - We interpret it the same way, where a smaller AIC value indicates a better model when comparing two nested models.

- We can use Likelihood-Ratio Tests to compare a restricted model with a unrestricted model.
 - Recall that the restricted model is the one with the restrictions required under the null (aka the "smaller" model).
- This works the same as the logit models we used the previous week.

Marginal effects

- Recall that marginal effects measure how a small change in an independent variable X influences the dependent variable.
- We can also compute marginal effects for Poisson and negative binomial models.

- Marginal effects measure the expected change in the dependent variable when an independent variable X increases by one unit, *ceteris paribus*.
- Similarly, we can compute average marginal effects, marginal effects at the mean, marginal effects at representative values and so on.

Example: Olzak & Shanahan's
(2014) work on group threat and
incarceration.

Did threats to white dominance raise the incarceration of African Americans and foreign-born whites?

More specifically: in U.S. cities from 1890 to 1910, what city characteristics predict how many members of each group were jailed for misdemeanors?

- Sample: 71 large U.S. cities, observed in 1890, 1904, and 1910 (178 city-years).
- Key Dependent Variables, estimated separately:
 - No. of African Americans in the local jail for a misdemeanor.
 - No. of foreign-born whites in the local jail for a misdemeanor.
- These are counts. Some cities have zero prisoners from a given group, so a rate is an awkward outcome.

- Key Independent Variables:
 - Occupational segregation (an isolation index)
 - Ratio of the group's paupers to native-born white paupers
 - Illiteracy
 - No. of attacks on the group in the previous decade

- They first compare a Poisson model with a negative binomial model.
- The dispersion parameter is positive and significant for both groups, so the variance is larger than the mean.
- They therefore report negative binomial estimates.
- The same cities appear more than once, so the models include a random effect for the city. We still read $\hat{\beta}$ as a factor of change.

Table 1: Negative binomial estimates, Model 4

	African Americans	Foreign-born whites
Occupational segregation	-2.052** (0.656)	-0.751 (0.515)
Pauper ratio	0.014 (0.011)	0.016** (0.005)
Prior attacks	0.002 (0.060)	0.289*** (0.064)
Observations	178	178

Coefficients are on the log-count scale. Standard errors in parentheses. ** $p < .01$,

*** $p < .001$.

Interpreting results

In column 2, the coefficient on occupational segregation of African Americans is -2.164 ($p < 0.01$).

- A one-unit increase would multiply the expected count by a factor of $\exp(-2.164) \approx 0.11$.
- The segregation index lies between 0 and 1, so a one-unit change is the whole scale. They instead report a one-standard-deviation change.
- e.g. “A one standard deviation decrease in occupational segregation is expected to cause a 47.5% increase in the number of African American prisoners ($p < 0.01$), *ceteris paribus*.”

The same threat does not operate the same way for both groups.

- For African Americans, incarceration rises where occupational segregation falls: job competition, not poverty.
- For foreign-born whites, occupational segregation is not significant. A one-standard-deviation increase in the pauper ratio is expected to cause a 27% increase in the number of foreign-born white prisoners, *ceteris paribus*.

Thanks for listening :)